```
import pandas as pd
```

```
df=pd.read_csv('/content/Salary_dataset.csv')
```

```
df.head()
```

	Unnamed: 0	YearsExperience	Salary
0	0	1.2	39344.0
1	1	1.4	46206.0
2	2	1.6	37732.0
3	3	2.1	43526.0
4	4	2.3	39892.0

```
df.tail()
```

	Unnamed: 0	YearsExperience	Salary
25	25	9.1	105583.0
26	26	9.6	116970.0
27	27	9.7	112636.0
28	28	10.4	122392.0
29	29	10.6	121873.0

```
x=df['YearsExperience'].values
```

```
y=df['Salary'].values
```

```
import numpy as np
x=np.array(x)
```

```
y=np.array(y)
```

```
x=X.reshape(-1, 1)
x
```

```
array([[ 1.2],
       [ 1.4],
       [ 1.6],
       [ 2.1],
       [ 2.3],
       [ 3. ],
       [ 3.1],
       [ 3.3],
       [ 3.3],
       [ 3.8],
       [ 4. ],
       [ 4.1],
       [ 4.1],
       [ 4.2],
       [ 4.6],
       [ 5. ],
       [ 5.2],
       [ 5.4],
       [ 6. ],
       [ 6.1],
       [ 6.9],
       [ 7.2],
       [ 8. ],
       [ 8.3],
       [ 8.8],
       [ 9.1],
       [ 9.6],
       [ 9.7],
       [10.4],
       [10.6]])
```

```
y=y.reshape(-1, 1)
y
```

```
array([[ 39344.],
       [ 46206.],
       [ 37732.],
       [ 43526.],
       [ 39892.],
       [ 56643.],
       [ 60151.],
       [ 54446.],
       [ 64446.],
       [ 57190.],
       [ 63219.],
       [ 55795.],
       [ 56958.],
       [ 57082.],
       [ 61112.],
       [ 67939.],
       [ 66030.],
       [ 83089.],
       [ 81364.],
       [ 93941.],
       [ 91739.],
       [ 98274.],
       [101303.],
       [113813.],
       [109432.],
       [105583.],
       [116970.],
       [112636.],
       [122392.],
       [121873.]])
```

```
from sklearn.model_selection import train_test_split
```

```
x_train, x_test, y_train, y_test=train_test_split(
    x, y,
    test_size=0.2,
    random_state=42
)
```

```
x_train
```

```
array([[10.4],
       [ 8.8],
       [ 4.1],
       [ 1.2],
       [ 2.3],
       [ 5.2],
       [ 3. ],
       [ 4.2],
       [ 4.1],
       [ 8. ],
       [ 1.4],
       [ 1.6],
       [ 9.1],
       [ 2.1],
       [ 7.2],
       [ 9.6],
       [ 6. ],
       [10.6],
       [ 6.9],
       [ 3.3],
       [ 4. ],
       [ 4.6],
       [ 6.1],
       [ 3.1]])
```

```
x_train.shape,x_test.shape, y_train.shape, y_test.shape
```

```
((24, 1), (6, 1), (24, 1), (6, 1))
```

```
from sklearn.linear_model import LinearRegression
```

```
model=LinearRegression()
```

```
model.fit(x_train, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
y_pred=model.predict(x_test)
```

```
print("actual salary:",y_test[:5])
print("predicted salary:",y_pred[:5])
```

```
actual salary: [[112636.]
 [ 67939.]
 [113813.]
 [ 83089.]
 [ 64446.]]
predicted salary: [[115791.21011287]
 [ 71499.27809463]
 [102597.86866063]
 [ 75268.80422384]
 [ 55478.79204548]]
```

```
from sklearn.metrics import mean_squared_error
```

```
mse=mean_squared_error(y_test, y_pred)
print("mean squared error:",mse)
```

```
mean squared error: 49830096.855908394
```

cal r^2

```
from sklearn.metrics import r2_score
```

```
R2=r2_score(y_test, y_pred)
print("R-squared value:", R2)
```

```
R-squared value: 0.9024461774180497
```

```
import matplotlib.pyplot as plt
```

```
plt.scatter(x_test, y_test)
plt.plot(x_test, y_pred)
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Salary vs Experience")
plt.show()
```

```
plt.scatter(x_test, y_test, label="Actual Salary")
plt.scatter(x_test, y_pred, label="Predicted Salary")

plt.xlabel("Actual salary")
plt.ylabel("Salary")
plt.title("actual vs predicted salary comparison")

plt.legend()
plt.show()
```

```
slope=model.coef_[0]
intercept=model.intercept_

print("Slope (m):", slope)
print("Intercept (c):", intercept)
```

```
Slope (m): [9423.81532303]
Intercept (c): [24380.20147947]
```

```
experience=[[5]]
```

```
predicted_salary=model.predict(experience)
print("predicted salary for 5 years of experience:", predicted_salary)
```

```
predicted salary for 5 years of experience: [[71499.27809463]]
```

```
predicted_salary = slope * 5 + intercept
print("predicted salary for 5 years of experience:", predicted_salary)
```

```
predicted salary for 5 years of experience: [71499.27809463]
```

```
print("actual salary (sample):", y_test[:5])
print("predicted salary (sample):", y_pred[:5])
```

```
actual salary (sample): [[112636.]
 [ 67939.]
 [113813.]
 [ 83089.]
 [ 64446.]]
predicted salary (sample): [[115791.21011287]
 [ 71499.27809463]
 [102597.86866063]
 [ 75268.80422384]
 [ 55478.79204548]]
```